

EXPERIENCE

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| <b>Software and AI Intern</b><br>MS Solutions Group   SPARK payments platform                                                                                                                                                                                                                                                                                                                                                                                                                                                                            | Summer 2026 |
| <ul style="list-style-type: none"><li>Engineered real-time payment-transaction monitoring features using <b>FastAPI, WebSockets, and Redis</b>, including regional and top-store views, financial indicators, and fraud alerts.</li><li>Implemented backend APIs and live data flows supporting user-facing transaction visualizations and fraud analytics.</li><li>Integrated a Gemini API chatbot for conversational access to dashboard data and company information in a payments environment serving 300+ customers across 30+ countries.</li></ul> |             |
| <b>Software Intern</b><br>Telnet                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                         | 2024-2025   |
| <ul style="list-style-type: none"><li>Developed and deployed a Docker-based live metrics dashboard from supervisor specifications.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                             |             |
| <b>Web Development Intern</b><br>Shamash IT                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              | 2023-2024   |
| <ul style="list-style-type: none"><li>Developed user-facing web interfaces with HTML, CSS, and JavaScript.</li></ul>                                                                                                                                                                                                                                                                                                                                                                                                                                     |             |

EDUCATION

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| <b>ESPRIT - Engineering Degree in Computer Science, Artificial Intelligence</b><br>Ranked <b>1st</b> in annual class ranking (2024-2025); 2nd (2023-2024). Final-year capstone (PFE) begins Fall 2027. | Expected 2028 |
| <b>Baccalaureate in Mathematics, Lycee El Menzah 9</b>                                                                                                                                                 | 2023          |

SELECTED PROJECTS

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| <b>Plant DNA Optimization and Expression Prediction</b><br>Deep Learning   Streamlit                                                                                                                                                                                                                                                                                              | 2025-2026      |
| <ul style="list-style-type: none"><li>Built a Streamlit application for plant genetic-data analysis and expression prediction; selected for the Bal des Projets deep-learning showcase.</li><li>Extended genomic analysis with host-aware codon optimization and grouped surrogate-model evaluation of simulated expression targets.</li></ul>                                    |                |
| <b>Reward Goblin - Reinforcement Learning Lab</b><br>Python   PyTorch   Gymnasium   Stable-Baselines3   Pymunk   FastAPI                                                                                                                                                                                                                                                          | 2026 - ongoing |
| <ul style="list-style-type: none"><li>Built a physics playground that exposes reward hacking by comparing trained PPO policies with an intended-task reference controller.</li><li>Evaluated reward versions across 3 training seeds, held-out starts and unseen layouts; separated task success from reward score and added heuristic exploit diagnostics.</li></ul>             |                |
| <b>Hydra - AI Coding Agent Orchestrator</b><br>TypeScript   Electron   Node.js   MCP   Claude Code   Codex                                                                                                                                                                                                                                                                        | 2026 - ongoing |
| <ul style="list-style-type: none"><li>Built a desktop workspace with MCP tools to spawn, prompt, monitor and restart coding agents; added resumable sessions, headless runs and mobile remote control.</li><li>Organized development around PLAN &gt; IMPLEMENT &gt; REVIEW &gt; FIX &gt; VERIFY &gt; DEMO &gt; APPROVAL, with controlled execution and human approval.</li></ul> |                |
| <b>Telekinesis CV - Gesture-Controlled Computer Vision</b><br>Python   OpenCV   MediaPipe   EdgeSAM   ONNX Runtime                                                                                                                                                                                                                                                                | 2026 - ongoing |
| <ul style="list-style-type: none"><li>Built a CPU-only webcam application combining hand tracking and point-prompted segmentation to select and manipulate real-object appearances in video.</li><li>Implemented pinch-to-grab, throwing, two-hand scaling and rotation, with background reconstruction and occlusion-aware compositing.</li></ul>                                |                |
| <b>LogiXpress - Sustainable Delivery Platform</b><br>PHP   MySQL   MVC                                                                                                                                                                                                                                                                                                            | 2024-2025      |
| <ul style="list-style-type: none"><li>Developed an MVC delivery application with client, order, and database-backed workflows; won a Bal des Projets award.</li></ul>                                                                                                                                                                                                             |                |
| <b>WaveWorx - Operations Desktop Application</b><br>C++   Qt   Oracle                                                                                                                                                                                                                                                                                                             | 2024-2025      |
| <ul style="list-style-type: none"><li>Implemented desktop modules for inventory and business records, including search, stock alerts, charts, and PDF reporting.</li></ul>                                                                                                                                                                                                        |                |

TECHNICAL SKILLS

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| <b>Languages:</b> Python, Java, C/C++, JavaScript, TypeScript, PHP, SQL                                                |
| <b>Software / Backend:</b> FastAPI, REST APIs, WebSockets, Redis, Django, Streamlit, HTML/CSS, Qt, MVC                 |
| <b>Data:</b> PostgreSQL, MySQL, MongoDB, Oracle, SQL / NoSQL                                                           |
| <b>Infrastructure / Tools:</b> Docker, Linux, Git, Node.js, Electron, MCP                                              |
| <b>AI / ML:</b> Machine Learning, Deep Learning, Generative AI, Gemini API, PyTorch, PPO, Gymnasium, OpenCV, MediaPipe |

CERTIFICATIONS & RECOGNITION

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| <ul style="list-style-type: none"><li>NVIDIA certificates in Machine Learning, Deep Learning, and Generative AI.</li><li>Bal des Projets: winner for LogiXpress; selected for deep-learning, C++ application, and Linux game showcases.</li></ul> |
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